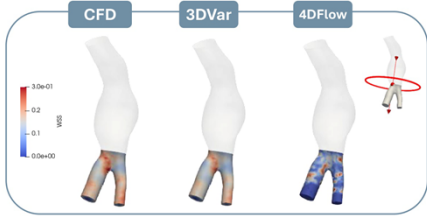
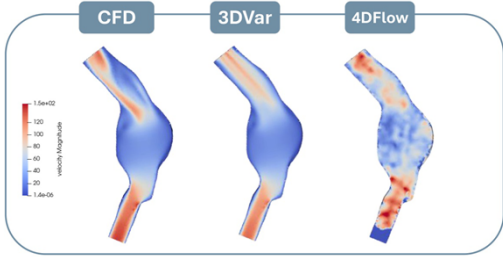


Alternative Texts for Your Images, Please Check and Correct them if Required

Page no	Fig/Photo	Thumbnail	Alt-text Description
5	fig1.jpg	The flow chart illustrates an optimization process for a discrete problem using an incremental pressure correction scheme. It includes mathematical equations for tentative velocity, pressure correction, and velocity update, with methods like Crank-Nicolson and Explicit Adam-Bashforth. It also shows the objective function, Tikhonov regularization term, and experimental measurements. The process involves the FEniCS Project and dolfin-adjoint, leading to adjoint equations and an optimization algorithm. A decision diamond checks if the number of minimization iterations is less than 120, leading to a VarDA result.	Flow chart illustrating an optimization process for a discrete problem using an incremental pressure correction scheme. The chart includes mathematical equations for tentative velocity, pressure correction, and velocity update, with methods like Crank-Nicolson and Adam-Bashforth. An objective function and Tikhonov regularization term are shown, involving variables u_n, d_n, and $T_n u$. Experimental measurements are depicted alongside. The process involves the FEniCS Project and dolfin-adjoint, leading to adjoint equations and an optimization algorithm. A decision diamond checks if the number of minimization iterations is less than 120, leading to a VarDA result.
8	fig1.jpg	The image is a panel figure composed of heatmaps showing velocity magnitudes and velocity difference magnitudes under different conditions. Each subfigure is labeled with parameters: α, β, R, and $J+R$, with values such as 10^{-3}, 10^{-1}, and 10^1. The heatmaps use a color gradient from blue to red, indicating varying magnitudes. The top row displays velocity magnitudes, while the bottom row shows velocity difference magnitudes. Color bars on the left provide scales for interpretation.	The image is a panel figure composed of heatmaps showing velocity magnitudes and velocity difference magnitudes under different conditions. Each subfigure is labeled with parameters: α, β, R, and $J+R$, with values such as 10^{-3}, 10^{-1}, and 10^1. The heatmaps use a color gradient from blue to red, indicating varying magnitudes. The top row displays velocity magnitudes, while the bottom row shows velocity difference magnitudes. Color bars on the left provide scales for interpretation.

Page no	Fig/Photo	Thumbnail	Alt-text Description
9	fig2.jpg		The image displays three 3D models of a bifurcated structure, each labeled as CFD, 3DVar, and 4DFlow. The models show variations in wall shear stress (WSS) represented by a color gradient from blue to red, with a scale on the left indicating values from 0.00e+00 to 3.0e-01. The models highlight differences in stress distribution across the lower sections. An inset on the right shows a smaller version of the structure with rotational and directional arrows, indicating movement or flow direction.
9	fig1.jpg		Three contour maps comparing velocity magnitude in a fluid flow simulation. Each map represents a different method: CFD, 3DVar, and 4DFlow. The maps show a tubular structure with varying colors from blue to red, indicating low to high velocity magnitudes. A color scale on the left ranges from 1.4e-06 to 1.5e+02.